New Tattoo Ink Can Vanish and Come Back

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Most tattoos stay the same forever. But **engineers** at the University of Colorado Boulder are making a new ink that can disappear and then come back.

The ink changes because of light. When **ultraviolet**, or UV, light hits it, the color fades until you can no longer see it. When normal light hits it later, the color returns. It works a bit like glasses that get darker in the sun.

The ink comes in three colors: yellow, blue and **magenta**. Each color is made of tiny beads. Inside the beads are special **chemicals** that react to light.

One of the inventors, scientist Carson Bruns, tested the ink on his own leg over about 12 months. All the colors worked as planned, and the ink kept working for more than two years.

The team hopes doctors might one day use it to make **temporary** marks on skin. For now, the ink is only for lab use until more safety tests are done.

Word Treasure

engineers -- People who use science and math to design and build new things.

ultraviolet -- A kind of light from the sun that our eyes cannot see.

magenta -- A bright purplish-pink color.

chemicals -- Substances that can change or react when mixed or hit by light or heat.

temporary -- Lasting only for a short time, not forever.

Background you need

A tattoo is a picture made with ink placed just under the skin.

Some materials change color when sunlight or special light hits them.

New products must be tested to make sure they are safe before people use them.

Break it down

WHO: Engineers at the University of Colorado Boulder, including Carson Bruns
WHAT: They made tattoo ink that can vanish and come back
WHERE: In their lab and on Carson Bruns' own leg
WHY: To make tattoos that change with light and maybe help doctors make temporary marks

Why it matters

It shows how science can make materials that change with light, like glasses that get darker in the sun.

Test what you remember

1. What is special about the new tattoo ink?

- (A) It smells like flowers
- (B) It glows like a flashlight
- (C) It never dries
- (D) It can disappear and come back

2. What makes the color fade?

- (A) Ultraviolet, or UV, light
- (B) Cold water
- (C) Wind
- (D) Loud music

3. Which colors does the ink come in?

- (A) Yellow, blue and magenta
- (B) Black and white
- (C) Red and green
- (D) Orange only

4. Who tested the ink on his own leg?

- (A) Carson Bruns
- (B) Haochen Wang
- (C) Dillon Amaya
- (D) Asaph Hall

5. How long did the ink keep working?

- (A) One week
- (B) One hour
- (C) More than two years
- (D) One day

6. Can anyone get this tattoo now?

- (A) No, it is only for lab use until more safety tests are done
- (B) Yes, at any shop
- (C) Only kids can
- (D) Only doctors can

Answer key (try the quiz first!): 1: D 2: A 3: A 4: A 5: C 6: A

Questions to think about

MY THOUGHTS (at least 20 words):